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RESEARCH ARTICLE

Research on Noise Reduction Method for Leakage Signal of Water Supply Pipeline

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ABSTRACT The negative pressure wave (NPW) signal caused by leakage of water supply pipeline contains a lot of noise due to the influence of water flow state and circuit noise, which will adversely affect the accuracy of leakage detection and leak location. To solve this problem, a novel pipeline pressure signal denoising method based on variational mode decomposition (VMD) is proposed in the paper. Firstly, the key parameters which affect the performance of VMD method: IMFs K and penalty factor α are optimized by genetic algorithm (GA) without prior knowledge. At the same time, different parameters are combined to form different reconstructed signals based on the correlation coefficient (CC). Secondly, the optimal combination of parameters is adaptively determined by using a fitness evaluation function based on the information entropy (IE). Thirdly, based on the optimal parameter combination, reconstruct the signal to achieve noise reduction. Finally, the denoised signal will be used for leak detection and leak location calculation. In order to verify the effectiveness of the proposed method, simulation analysis and laboratory experiments are carried out. Simulation analysis results show that the noise suppression effect of our method is better than that of wavelet, empirical mode decomposition (EMD) and VMD. The highest SNR obtained by our method is 16.187. The laboratory experiment results showed that the NPW signal denoised by proposed method had less error and more stable results in location, with an average relative error of 2.97%.

INDEX TERMS Negative pressure wave, leak location, variational mode decomposition, genetic algorithm, correlation coefficient, information entropy.

I. INTRODUCTION

Urban water supply pipelines provide a continuous transport of residential water, can be operated at very low labor costs, and can also provide reasonable protection against adverse weather and environmental conditions. The most common problems that challenge the sustainable and safe operation of any pressurized pipeline are leaks and bursts [1]. The leakage of the water supply system will not only lead to the waste of water resources and the inefficient distribution of energy in the entire pipe network, but also under low pressure conditions, pollutants may invade the pipe network and pollute the water quality [2]. In order to solve the above problems, various leak detection and leak location methods have been proposed, including mass-volume balance, transient modeling, distributed fiber, acoustic emission, and NPW [3], [4]. Among them, the location method based on the principle of NPW has been widely used because of its advantages of simple deployment, low cost, high sensitivity, strong remote applicability and strong robustness [5]. Therefore, based on the principle of NPW, this paper will study the method of leakage detection and location of water supply pipeline.

When there is a leak in the water supply pipeline, the pressure will drop rapidly at the leak point, and the pressure drop will spread to the upstream and downstream of the pipeline at a certain speed, thus forming a NPW. The velocity of the NPW is approximately constant when the pipeline conditions and the surrounding environment are the same. There is a

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certain difference in the time for the NPW to reach the sensor at different positions on the pipeline. The leak point can be located by the NPW speed, the distance between the sensors and the time difference. However, the NPW signal will be mixed with a large amount of noise during the propagation process, and the noise will mask leak information carried by the NPW signal. Therefore, it is necessary to pre-process signal noise reduction [6], [7]. In order to more accurately extract the leak characteristics carried by the NPW signal, it is necessary to study a reasonable pipeline pressure signal processing method to reduce the noise to obtain the purest possible NPW signal [8].

New signal processing techniques have been proposed continuously in recent years. Benefiting from Fourier transform, signal processing techniques have been extended to the frequency domain and are widely used in fault diagnosis [9]. However, the Fourier transform does not have temporal resolution and usually analyses signals at the global level, so it is not applicable in every case, especially when local features of the signal need to be analyzed. The wavelet analysis method [10] compensates for some of the shortcomings of the Fourier transform. It has better localization properties in both the time and frequency domains, but the selection of the wavelet basis function has a great impact on the analysis results, and if a suitable wavelet basis function cannot be selected, some features of the signal will be lost. EMD [11] is a new method proposed by Huang in 1998 to deal with non-stationary signals, which is widely used in fault diagnosis [12]. It can decompose the signal into a series of IMFs, but the EMD does not have a complete mathematical basis, which leads to problems such as over-decomposition, modal mixing and endpoint effects in its engineering applications [13]. To solve the problems of over-decomposition and modal mixing in EMD, Huang introduced white noise into EMD and proposed the EEMD method [14]. Methods such as mirror extension are used to solve the endpoint effect problem of EMD [15], [16]. Based on EMD, LU et al. proposed SNR-EMD to deal with pipeline pressure signals [17], but the method still relies on methods such as mirror extension to overcome a series of drawbacks of EMD. Local mean decomposition (LMD) outperforms EMD in terms of suppressing endpoint effects and reducing the number of iterations, but it still suffers from endpoint effects and the signal will advance or lag when the number of smoothing is hig [18]. Dragomiretskiy et al. in 2014 proposed a blind separation method for non-stationary signals known as VMD [19], which constructs a variational model and introduces penalty factors and Lagrange operators to transform the constrained variational problem into an unconstrained variational problem before iteratively solving it. VMD overcomes almost all the shortcomings of EMD and has been widely used in pattern recognition and fault diagnosis in recent years. However, VMD is not as adaptive as EMD, it requires pre-set parameters, and the decomposition effect is limited by two parameters: the number of IMFs (K) and the penalty factor (α) [20].

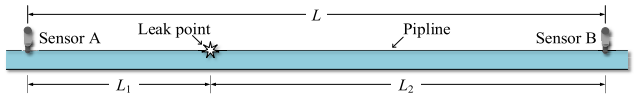

FIGURE 1. Principle of NPW.

Aiming at the problems of existing methods in processing pipeline pressure signals, taking into account that the signal components obtained by VMD have frequency domain sparsity, based on VMD, a novel adaptive denoising method for pressure signal of water supply pipeline is proposed in this paper.

Firstly, the global optimization of two key factors in VMD: IMFs (K) and penalty factor (α) is realized by using GA without prior knowledge. Secondly, based on the CC, the reconstructed signals under different parameter combinations are completed. Moreover, the optimal combination of parameters is adaptively determined by using a fitness evaluation function based on IE. Thirdly, based on the optimal parameter combination, reconstruct the signal to achieve noise reduction. Finally, the denoised signal will be used for leak detection and leak location calculation. The structure of the full text is arranged as follows.

Chapter I introduces the research background of this paper. In Chapter II, some theories and methods involved in this paper are introduced. The simulation experiments are presented in Chapter III. In Chapter IV, an experimental environment is established in the laboratory to validate the effectiveness of proposed method by comparing it with other methods. Finally, conclusions are outlined in Chapter V.

II. METHODOLOGY

A. PRINCIPLE OF LEAK LOCATION BASED ON NPW

The principle of water supply pipeline leak location based on NPW is shown in Figure 1. Assuming that the leak point is located between two pressure sensors A and B, when the leak occurs, the leak point can be located by using the time difference between the NPW reaching two sensors, combined with the distance between pressure sensors and the velocity of the NPW. The time difference between the NPW and the two sensors is obtained by the time difference between the two NPW signal turning points. As shown in Figure 1, the distance between sensors is L. The distance between the leak points and sensors A and B is denoted as L_1 and L_2, respectively.

The time required for the NPW signal generated by leak to reach sensors A and B is denoted as t_1 and t_2, respectively. $t_1 = L_1/v$, $t_2 = L_2/v$.

$$L_1 = \frac{1}{2}(L + v\Delta t) \tag{1}$$

where v is the NPW speed and the time difference is $\Delta t = t_1 - t_2$. Therefore, the key factor influencing the accuracy of leak location is the time difference Δt, which is obtained from the turning point of the NPW. However, noise can weaken the characteristics of the turning point, leading to significant errors in the location results. In order to improve the accuracy

of turning point extraction, this paper will investigate denoising methods for pipeline pressure signals.

B. VMD

VMD is a fully non-recursive decomposition model commonly used for blind signal separation which involves the construction and solution of constrained variational problems. It decomposes the signal into K amplitude modulation frequency (AM-FM) signals [21] as follows

$$u_k(t) = A_k(t)\cos(\phi_k(t)), k = 1, 2, \cdots, \mathrm{K} \tag{2}$$

where the phase $\phi_k(t)$ is a non-decreasing function, i.e. $\phi_k'(t) \geq 0$. The envelope function $A_k(t)$ and the instantaneous frequency $\phi_k'(t)$ vary much more slowly than the phase $\phi_k(t)$. On a sufficiently long-time scale $[t-\tau, t+\tau](\tau \approx 2\pi/\phi_k'(t))$, the IMFs can be seen as purely harmonic signals with amplitude $A_k(t)$ and instantaneous frequency $\phi_k'(t)$, i.e., the bandwidth of the modes is limited.

In order to satisfy that the sum of the estimated bandwidths of each IMF is minimum and equal to the input signal, the following constrained variational model is constructed

$$\min_{\{u_k\},\{\omega_k\}} \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\}, \quad s.t. \sum_k u_k = f \tag{3}$$

where $\{u_k\}$ denotes each IMF, $\{\omega_k\}$ is the center frequency of each IMF, $(\delta(t) + j/\pi t)u_k(t)$ denotes the spectrum obtained by transforming the $u_k(t)$ by the Hilbert transform, f denotes the original signal. By searching for the optimal solution to the above variational problem, the VMD method iteratively updates the center frequency and bandwidth of each component and adaptively decomposes the IMF components. To transform the constructed constrained variational problem into an unconstrained variational problem for solution, the penalty factor α and the Lagrange operator $\lambda(t)$ are introduced, and the Lagrange expression is extended as follows

$$\begin{aligned} &\Gamma(\{u_k\}, \{\omega_k\}, \lambda) \\ &:= \alpha \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ &\quad + \left\| f(t) - \sum_k u_k(t) \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_k u_k(t) \right\rangle \end{aligned} \tag{4}$$

The penalty factor α reduces the distortion of the reconstructed signal in the presence of Gaussian noise, and the Lagrange operator $\lambda(t)$ guarantees tight constraints. The alternate direction method of multipliers (ADMM) is used to solve the unconstrained variational problem of (4). The search for the "saddle point" of the widening Lagrange function is performed by alternately updating u_k^{n+1}, ω_k^{n+1} and λ^{n+1}, where n denotes the number of iterations to convergence in the solution, and the iterative steps are as follows:

Step 1. Initialize $\{u_k^1\}$, $\{\omega_k^1\}$, λ_k^1 and n to 0.

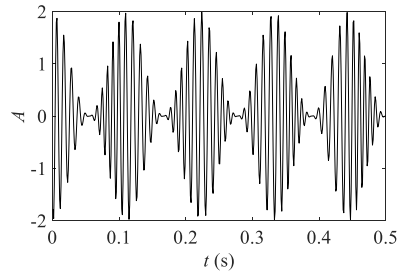

FIGURE 2. Time-domain waveform of the simulated signal s(*t*).

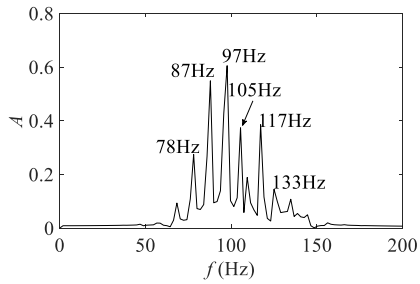

FIGURE 3. Spectrum of the simulated signal s(t).

Step 2. $n = n + 1$, entering the outer loop.

Step 3. Entering the inner loop.

Step 4. Update u_k and ω_k, end the inner loop when the number of decompositions reaches K, at which u_k and ω_k are determined.

Step 5. When accuracy $\varepsilon > \sum_k (\left\| u_k^{n+1} - u_k^n \right\|_2^2 / \left\| u_k^n \right\|_2^2)$, then ends the outer loop. Otherwise, go to step 2.

VMD obtains u_k and ω_k by the above iterations.

C. EFFECT OF PARAMETER SETTINGS ON VMD

The pipeline pressure signal measured by the pressure sensor contains a sizable AM-FM signal. Therefore, this subsection explores the effect of VMD parameters on the decomposition effect by simulating the AM-FM signal as an example, taking into account the diversity of the frequency band distribution, which is

$$s(t) = (1 + \cos(18\pi t))\cos(214\pi t + 2\cos(20\pi t)) \tag{5}$$

When the sampling frequency f_s is 500 Hz, the time and frequency domain diagrams of the simulated signal $s(t)$ are shown in Figure 2 and Figure 3 respectively, where A indicates the amplitude of the simulated signal, t indicates the sampling time and f indicates the frequency of the signal.

The frequency components of the simulated signal are mainly concentrated at 78 Hz, 87 Hz, 97 Hz, 105 Hz, 117 Hz and 133 Hz, as indicated by the labels in Figure 3. In addition to K and α, other parameters have little influence on VMD effect [22], so this section focuses on the influence of K and α on the calculation results of the algorithm.

1) THE EFFECT OF IMF QUANTITY K ON VMD

When analyzing the influence of K on the effect of VMD, the penalty factor is set to two times the sampling frequency f_s of

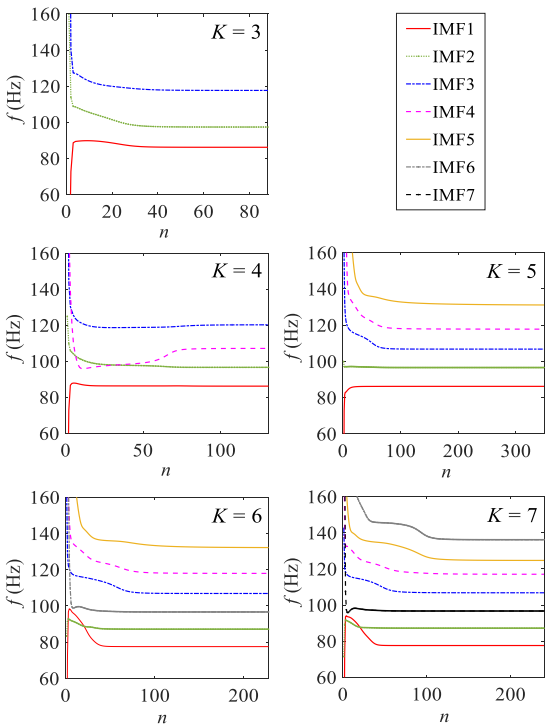

FIGURE 4. Variation of central frequency with IMFs number K.

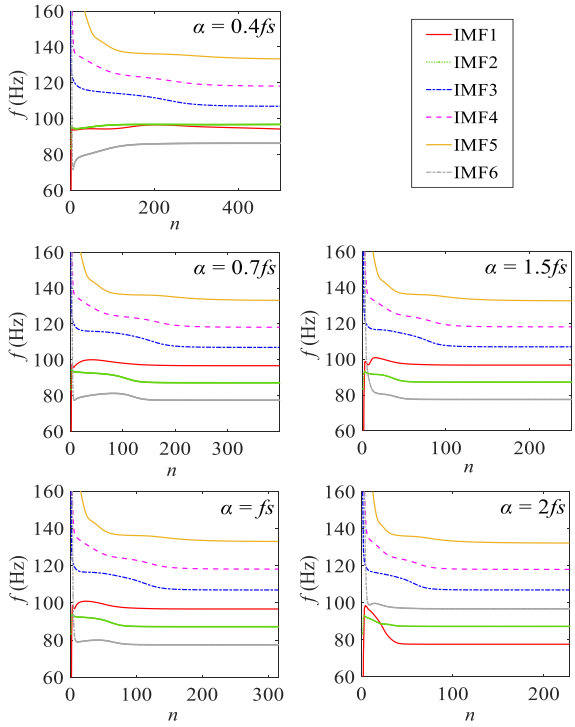

FIGURE 5. Variation of central frequency with penalty factor α.

the simulated signal $s(t)$ [23]. K is taken as 3, 4, 5, 6 and 7, respectively, and the simulated signal $s(t)$ is decomposed by the VMD. The Figure 4 shows the variation of the center frequency of each IMF during the decomposition, where n represents the number of iterations in the decomposition process.

As can be seen from Figure 4, when $K = 3$, the central frequencies of the three IMFs are separated with the iteration of VMD, but the frequency components of the decomposed signal are less than the six components contained in the simulated signal, which is in under-decomposition situation. At $K = 4$, the main frequency components are still not fully decomposed by the VMD. When $K = 5$, the five main frequency components contained in the simulated signal are accurately decomposed by the VMD, but the 78Hz frequency component is omitted. At $K = 6$, although the central frequencies of each IMF overlapped early in the VMD iteration, they are later separated and the full frequency component is decomposed. $K = 7$, the VMD produces a spurious component around 140 Hz. These show that, with a fixed penalty factor α, a suitable number of IMFs K is chosen to better decompose the main frequency components contained in the original signal, while an inappropriate choice will result in under- and over-decomposition.

2) EFFECT OF PENALTY FACTOR α ON VMD

According to the prior knowledge obtained from spectral analysis, the IMFs number K is set to 6 when studying the influence of α on VMD effect. Set α to $0.4f_s$, $0.7f_s$, f_s, $1.5f_s$ and $2f_s$ respectively. The variation of the center frequency of each IMF during the decomposition are shown in the Figure 5.

For $\alpha = 0.4f_s$, the frequency component at 78 Hz is not decomposed and a spurious component is generated at 97 Hz. For $\alpha = 0.7f_s, f_s$ and $1.5f_s$, the main frequency components are completely separated out, and the number of iterations completed by the decomposition decreases as α increases. When $\alpha = 2f_s$ the main frequency components are completely separated out, but the number of iterations increases compared to $\alpha = 1.5f_s$, indicating that the efficiency of the decomposition decreases as α increases. The analysis results show that when K is fixed, the selection of α has great influence on the decomposition effect and efficiency.

In summary, when utilizing VMD for blind signal separation, there is an irregular challenge in the selection of parameters K and α. Failure to choose appropriate parameters may result in inefficient decomposition and unsatisfactory outcomes.

The sequential optimization method used in this subsection can only yield locally optimal parameters. Given that GA have shown excellent efficacy and broad applicability in multi-objective nonlinear optimization problems [24], this paper employs a parallel search for optimal VMD parameters using GA in combination with CC and IE.

D. PROPOSED METHOD

1) SIGNAL RECONSTRUCTION

The pipeline pressure signal is a non-priori signal with unknown characteristics. When there is no fixed interference source around the pipeline, the noise has low or no

correlation with the original collected pressure signal [25]. At the same time, considering that the correlation coefficient (CC) describes the closeness of the relationship between two variables from a statistical point of view, it can reflect thel strength of the linear correlation between two variables [26]. CC is defined as (6).

$$CC = \frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\left\{\sum_{i=1}^{n}(X_i - \overline{X})^2 \sum_{i=1}^{n}(Y_i - \overline{Y})^2\right\}^{1/2}} \tag{6}$$

where n is the length of the signal, X and Y are signals, $\overline{X}$ and $\overline{Y}$ are the mean values of the n data in the signals.

Therefore, when VMD is used to decompose the original pressure signal into multiple IMFs, this paper uses CC between IMFs to original pressure signal to distinguish these IMFs into valid IMFs (components with pressure information) and invalid IMFs (components with noise), and reduces the noise in the signal by discarding the invalid IMFs and reconstructing the valid IMFs. If I valid IMFs are represented as $w_I(t)$, then the reconstruction signal $r(t)$ is

$$r(t) = \sum_{I} w_I(t) \tag{7}$$

2) FITNESS EVALUATION FUNCTION BASED ON INFORMATION ENTROPY

Entropy can reflect the state of a system, and IE can quantifies the noise amount in reconstruction signal. The higher the entropy value of a signal, the sparser and more uncertain it is, indicating more noise in the signal and a lower signal-to-noise ratio (SNR) (and vice versa) [27]. In this paper, the IE of the reconstructed signal is used as part of the fitness evaluation function. In recent years some entropy indicators have been successfully applied in the field of signal processing [28], [29], [30], considering the reasonable statistical properties of fuzzy entropy [31], this paper selects fuzzy entropy to evaluate the noise content in the signal.

If there is a signal of length n, it is decomposed into $i = n - m + 1$ sequences by window length m, which can be expressed as

$$X(i) = [x(i), x(i+1), \cdots, x(i+m-1)] - \sum_{j=0}^{m-1} x(i+j) \tag{8}$$

Calculate the distance between each sequence and all sequences d_{ij}.The sequence fuzzy membership is obtained from the distance d_{ij}

$$D_{ij}^m = \mu(d_{ij}^m, n, r) = \exp\left(\frac{-(d_{ij}^m)^n}{r}\right) \tag{9}$$

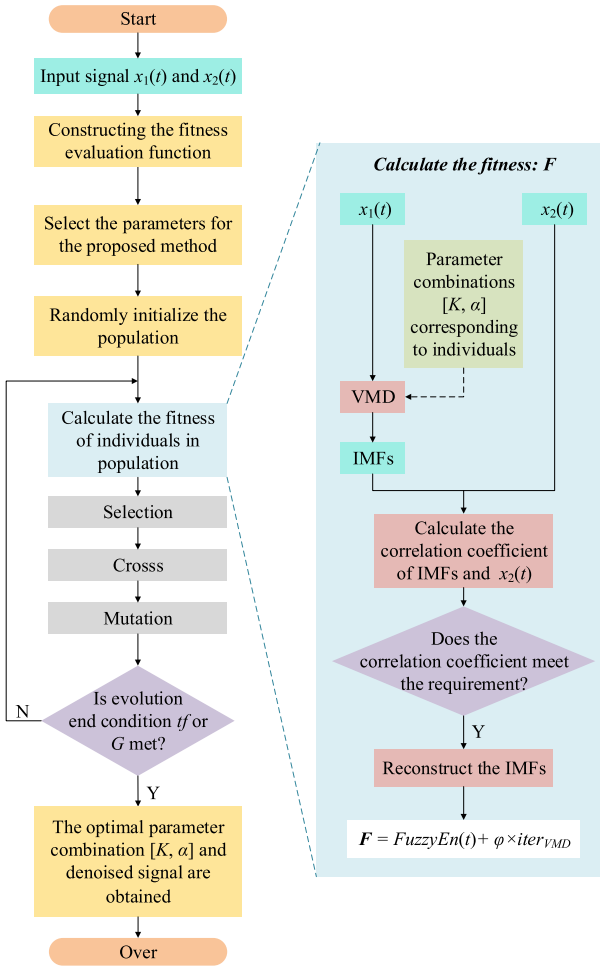

FIGURE 6. The proposed method.

where r is the similarity tolerance threshold. Next, the membership is averaged over all memberships except itself

$$\phi^m(n, r) = \frac{1}{N-M}\sum_{j=1}^{N-m}\left[\frac{1}{N-M-1}\sum_{j=1, j\neq i}^{N-M} D_{ij}^m\right] \tag{10}$$

The window length m is increased to $m+1$ and the above steps are repeated to obtain $\phi^{m+1}(n, r)$, and the fuzzy entropy $FuzzyEn(t)$ is calculated

$$FuzzyEn(t) = \ln\phi^m(t) - \ln\phi^{m+1}(t) \tag{11}$$

In the above method of reconstructing signals, different VMD parameters correspond to different reconstructed signals. In order to get the reconstruction signal with the lowest IE, this paper searches the VMD parameters by GA with the goal of the lowest IE of the reconstruction signal. The fitness evaluation function aims to quantitatively evaluate the fitness of each individual during the evolution of GA, and the goal of the method proposed in this paper is to obtain reconstruction signals with minimum IE. Therefore, after considering the search objective of GA and the decomposition efficiency of VMD, the adaptation evaluation function F is determined as

$$F = FuzzyEn(t) + \varphi \times iter_{VMD} \tag{12}$$

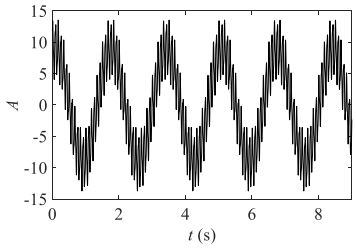

FIGURE 7. Time domain waveform of $s_1(t)$.

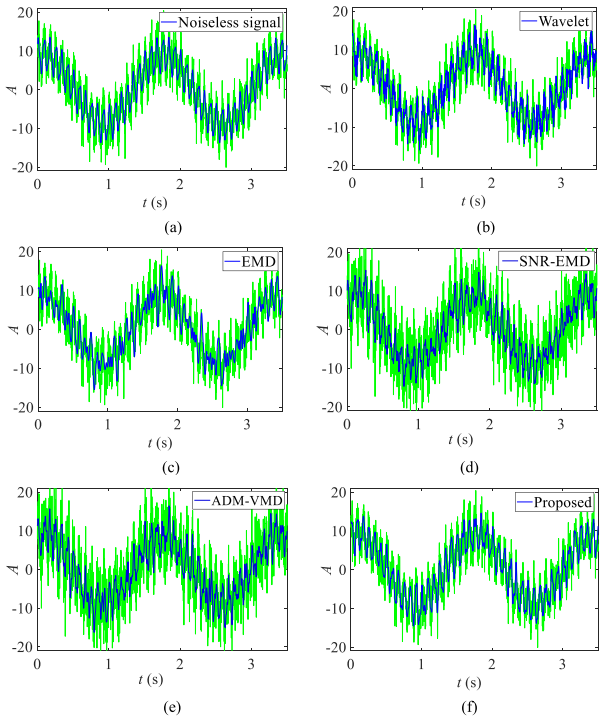

FIGURE 8. Comparison of noise reduction effect of several methods.

where φ is the empirical factor of the fitness evaluation function and $iter_{VMD}$ is the number of iterations at VMD.

It follows that if VMD is desired to obtain optimal decomposition parameters, the optimal individual within the population during optimization should have the best fitness min F in the current population. All the processes of our method in this paper are depicted in Figure 6, as follows:

Step1, sensors A and B installed upstream and downstream of the pipeline collect the pressure signals $x_1(t)$ and $x_2(t)$. Fitness evaluation function is constructed. Initialize the parameters of our method with an optimization range set to $[K_1, K_2]$ for the number of IMFs K and a search range set to $[\alpha_1, \alpha_2]$ for the penalty factor α. The signal to be reduced $x_1(t)$, the contrast signal $x_2(t)$ is fed into our method and the GA generates an initial population of size M (10 randomly generated combinations of parameters $[K, \alpha]$).

Step2, enter the cycle of GA evolution. The fitness of M individuals is calculated. The VMD is performed with the parameter combinations corresponding to M individuals in the current population to the noise reducing signal $x_1(t)$, and

TABLE 1. Comparison of SNR of several noise reduction methods.

SNR of noisy signal (dB)	Denoising methods	SNR of denoised signals (dB)	Improvement of SNR (dB)
	Wavelet	13.470	6.670
	EMD	9.820	3.020
6.800	SNR-EMD	14.181	7.381
	ADM-VMD	14.620	7.820
	Proposed	**16.187**	**9.387**

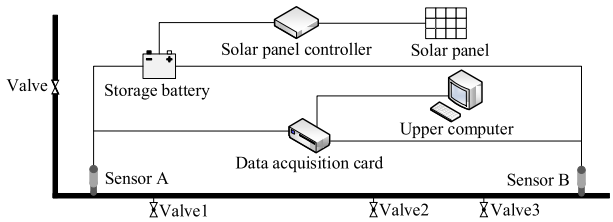

FIGURE 9. The experimental platform.

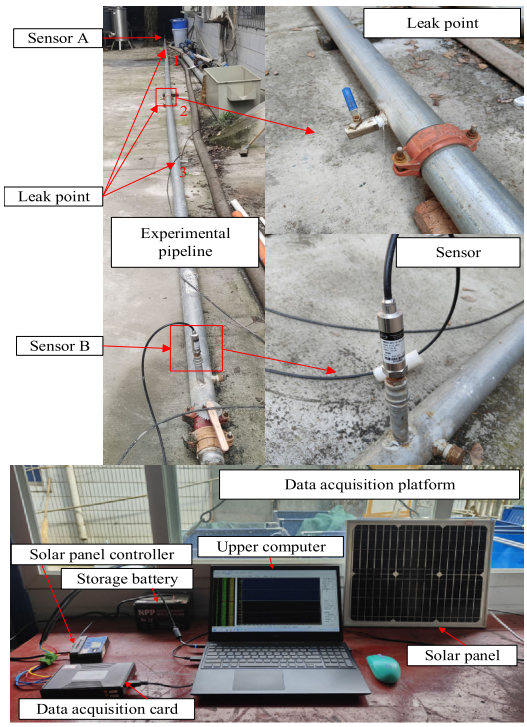

FIGURE 10. The site layout.

M groups of IMFs are obtained. Then, the IMFs in each group and contrast signal $x_2(t)$ are subjected to CC operation, and those whose CC are greater than meeting the requirements are valid IMFs, and the valid IMFs in each group are summed to obtain the reconstructed signal corresponding to M individuals. The fitness is obtained according to the IE of reconstructed signal and the number of VMD iterations.

Step3, a series of operations of the GA is performed. Roulette selection algorithm is used to select two individuals from the current population for the crossover operation (the

TABLE 2. Leak point distribution.

Pipeline length number	L (m)	Leak point number	L_1 (m)
		1	14.47
1	17.34	2	11.34
		3	2.47
		4	17.34
2	23.34	5	14.47
		6	2.47

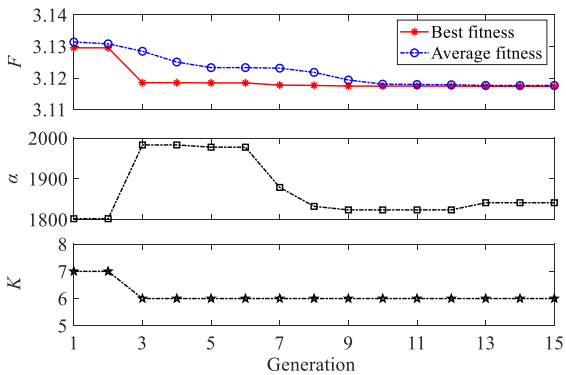

FIGURE 11. Evolutionary trends.

higher the fitness the higher the probability that the individual will be selected, the same individual may be selected more than once), and whether or not the crossover is performed depends on the crossover probability P_c. Then, a roulette selection algorithm is again used to randomly select one individual from the current population for the mutation operation (the selection rules are the same as for the crossover operation), and whether or not the mutation depends on the mutation probability P_m. After the crossover operation is executed M times, the mutation operation is also executed M times and the old population is replaced by the resulting new population.

Step4, the cycle of evolution stops when the end condition *tf* or *G* is met. After the G*th* execution of **Step 2**, the chromosome with the best fitness individual is decoded to obtain the optimal VMD parameters $[K, \alpha]$, and the reconstructed signal corresponding to the best fitness individual is the denoised signal output by our method.

In the above steps, P_c, P_m, M, G, and tf respectively represent crossover probability, mutation probability, population size, generation threshold, and individual fitness threshold [32].

III. SIMULATION ANALYSIS

Before denoising the real signal, the simulation signal is used to study the effectiveness of the algorithm. MATLAB is employed to generate the AM-FM noiseless signals $s_1(t)$ and $s_2(t)$ as outlined in (13). Consistent with the frequency of the acquisition card used in the experiment, the sampling frequency of the simulation signal is set to 500Hz and the

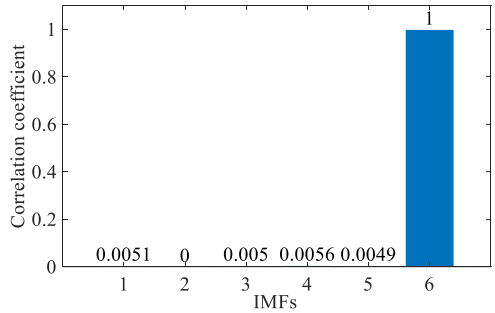

FIGURE 12. Correlation coefficient between each IMFs with the contrast signal.

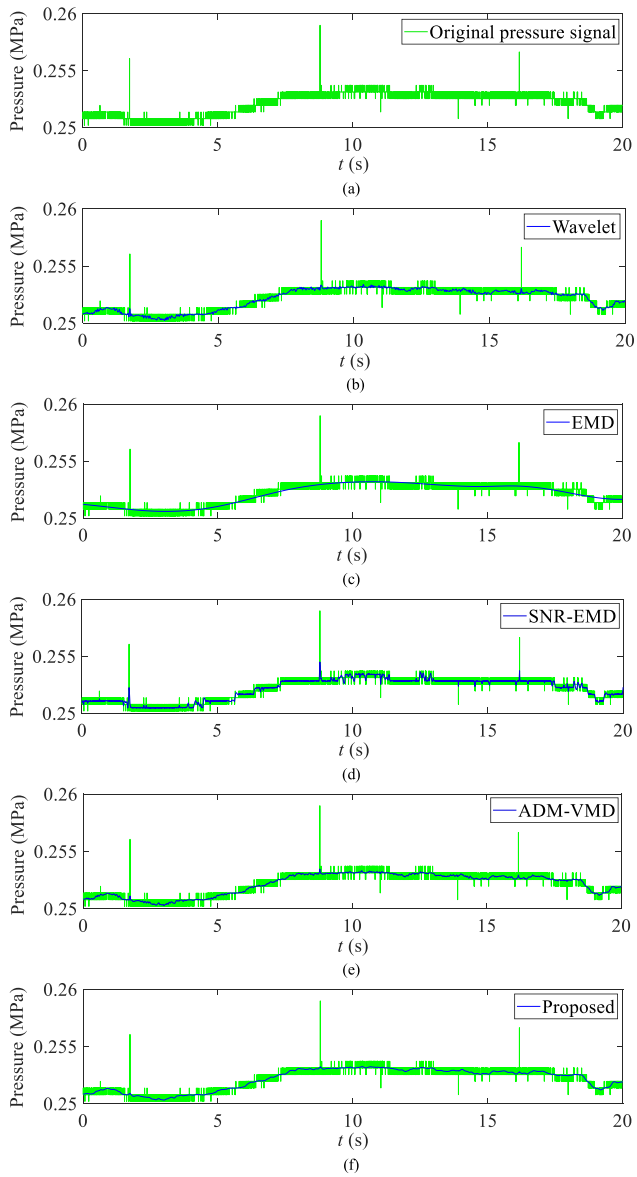

FIGURE 13. Comparison of denoising of pressure signals in the leaked-free state.

sampling time is 9s.

$$s_{1,2}(t) = \sin(540\pi) + 4\sin(216\pi + 0.7) + 9\sin(10.8\pi + 1.2) \quad (13)$$

The waveform of $s_1(t)$ is depicted in Figure7. The signal $s_2(t)$ serves as the contrast signal for $s_1(t)$, displaying a time-domain waveform similar to that of $s_1(t)$. Due to a time delay in the NPW signals received by the two sensors, the onset of the simulated signal $s_2(t)$ is delayed by 0.04s compared to $s_1(t)$.

A noisy signal with SNR of 6.800 dB is obtained by mixing Gaussian white noise into the signals $s_1(t)$ and $s_2(t)$. The noisy signals are denoised using wavelet, EMD, SNR-EMD [33], ADM-VMD [34] and our proposed method, respectively. Among them, the ADM-VMD method searches parameter K with the minimum sample entropy as the target by fixing parameter α as 2000. Figure 8 shows the simulation signal in the 3.5s range and the processing results of the noise simulation signal by four methods.

Figure 8 (a) is the simulation signal. The parameters for the wavelet are set as "sym4" wavelet and four levels of wavelet decomposition. The denoising results show that, as shown in Figure 8 (b), the denoised signal still contains a significant amount of noise. Figure 8 (c) shows the signal after undergoing EMD processing, and it can be observed that waveform distortion is quite severe. The signal after denoising by SNR-EMD is shown in Figure 8 (d), there is still a lot of noise in the signal, and there is obvious distortion at the 3s. Figure 8 (e) shows that the noise reduction capability of ADM-VMD is insufficient. The signal after being processed by our proposed method is shown in Figure 8 (f). The denoised signal is overall smooth, exhibiting the highest consistency with the noise-free signal.

At the same time, we use the signal-to-noise ratio to further characterize the denoising effect of various algorithms. The SNR is used to quantitatively evaluate the denoising effect of several methods, and the SNR of the noisy signal is calculated as in (14).

$$SNR = 20\lg(S/N) \tag{14}$$

where S is signal power and N is noise power.

The SNR results of each method are shown in Table 1. As can be seen from Table 1, the SNR of wavelet, EMD, SNR-EMD and ADM-VMD are 13.470, 9.820, 14.181 and 14.620, respectively. The SNR after denoising by our proposed method is 16.187. The proposed method have the best noise reduction effect, and the SNR improves by 9.387 compared with the noisy signal.

IV. EXPERIMENTAL STUDY

A. EXPERIMENTAL CONDITIONS

After testing the effect of the algorithm with simulation signals, we set up an experimental platform to further verify the denoising effect of various methods with real signals. The schematic diagram of the experimental platform is shown in Figure 9. The experimental equipment comprises the upper computer, data acquisition card, HM-90 high-frequency dynamic pressure sensors, the experimental water supply pipeline, solar panel, solar panel controller, and storage battery. The site layout and some equipment are shown in Figure 10.

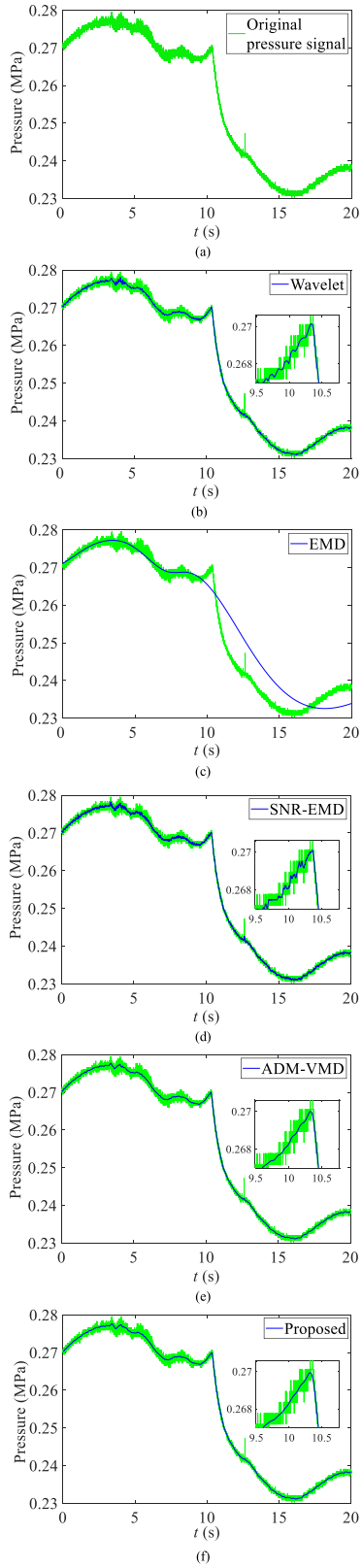

FIGURE 14. NPW signal denoising comparison.

TABLE 3. Comparison of information entropy of denoised signals.

IE of the original signal	Method	IE of the denoise signal	Decreased value
2.412	Wavelet	2.303	0.109
	SNR-EMD	2.313	0.099
	ADM-VMD	2.292	0.120
	Proposed	**2.281**	**0.131**

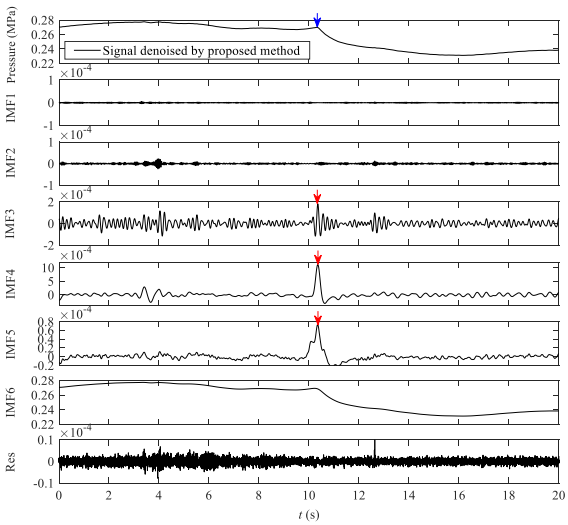

FIGURE 15. Denoised signal A′ and IMFs.

In this paper, two types of total pipeline lengths are established in the experiment, as detailed in Table 2. L represents the length along the pipeline between the two sensors, while L_1 denotes the actual distance from the simulated leak point to sensor A. The experimental pipeline consists of multiple sections of short pipes, allowing the total length L of the pipeline to be increased by adding more short pipes during the experiment. As the total pipeline length L is extended, the position of the simulated leak point relative to the sensor will correspondingly change. Consequently, there are a total of six numbered leak points. This paper simulates different leak areas by changing the valve opening.

In the experiment, a pipeline leak is created by opening a valve. The pipe diameter is 0.08m. The pipe wall thickness is 0.003m. The experience value of negative pressure wave velocity is 1000 m/s. The sampling rate is 500Hz.

B. NOISE REDUCTION

1) LEAK-FREE STATE

To validate the effectiveness of the proposed method in suppressing noise in pipeline pressure signals, the pressure signals from the pipeline in a leak-free state are collected. Specifically, we denote the pressure signal collected by sensor A upstream of the pipeline as $x_1(t)$ and the pressure signal collected by sensor B downstream as $x_2(t)$. Taking $x_1(t)$ as an example, we apply the proposed method to process the original pressure signal. The parameters P_c, P_m, M, G, and tf for our method are set to 0.8, 0.2, 10, 15, and 0.15, respectively. The Figure 11 illustrates the evolutionary

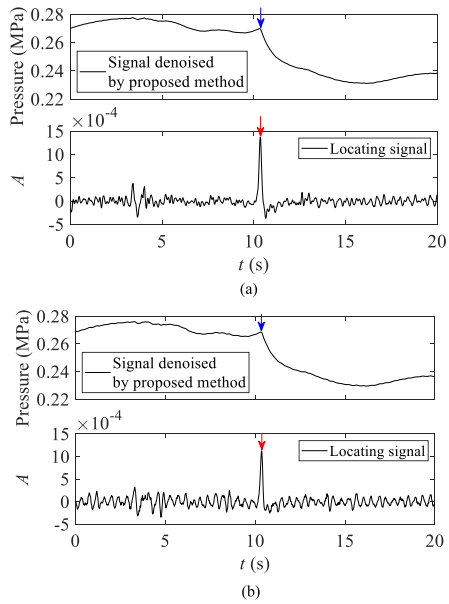

FIGURE 16. Denoised signals and locating signals.

process over 15 iterations. It can be observed that during the evolutionary process, the average fitness of the population and the best fitness curve exhibit a tendency to decrease together and converge, indicating the convergence of our method. There is essentially no change trend in the fitness of the best individuals from the 8th generation to the 15th generation, suggesting that the population has matured by the 8th generation of evolution. The final output of the optimal VMD parameters is $[K, \alpha] = [6,1842]$.

The CC during the reconstruction of $x_1(t)$ using the method proposed in Chapter II are illustrated in Figure 12. IMF6 is identified as a valid component, and therefore, it is chosen for reconstruction.

The denoised signals obtained through the proposed method and the other four methods are depicted in Figure 13.

As can be seen from the Figure 13 (a), the original pressure signal in the unleaked state has small unstable fluctuations, and a large amount of high frequency noise and random noise mask the details of the pressure fluctuations, which is very unfavorable to leak location.

It undergoes wavelet denoising as a first step (with parameter settings consistent with the simulation study in Chapter III), resulting in Figure 13 (b). As can be seen from Figure 13 (b), the signal after wavelet denoising shows an overall zigzag fluctuation, and some sudden interference cannot be effectively suppressed. Figure 13 (c) shows the de-noising effect of EMD method on pressure signal. As can be seen from the figure, the noise has been significantly suppressed. However, the signal is distorted at around 19s due to excessive processing. As can be seen from Figure 13 (d), there are still many strong disturbances in the signal after SNR-EMD processing. The signal processed by the ADM-VMD denoising method is shown in Figure 13 (e). After denoising,

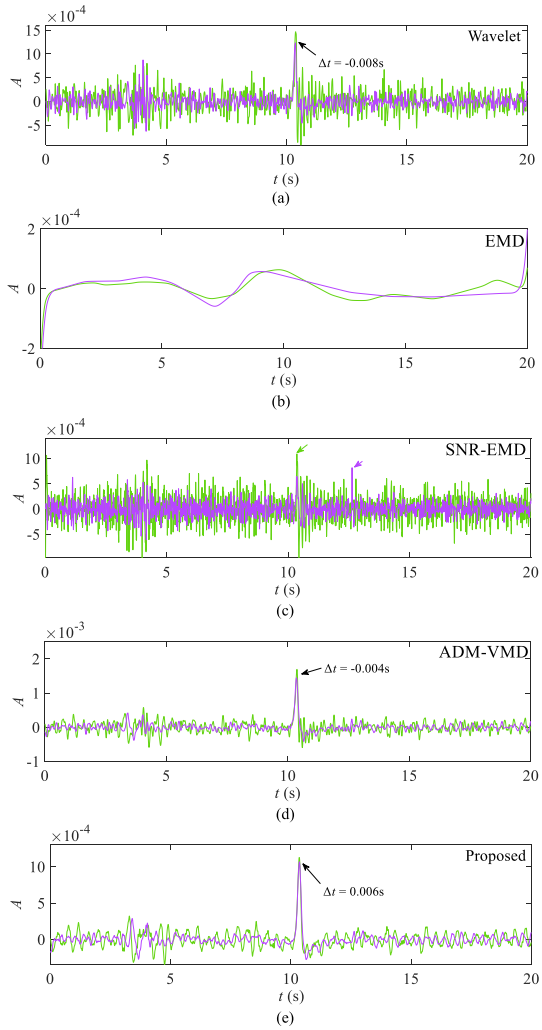

FIGURE 17. **Comparison of locating signals.**

the signal is smoother overall, but it does not completely filter out the three strong interferences.

The noise reduction effect of the proposed method in the text is shown in Figure 13 (f). It can be seen from the figure that the proposed method has demonstrated good noise suppression performance. It can effectively filter out noise interference while retaining the characteristics of pressure fluctuation.

2) LEAK STATE

In order to further evaluate the noise reduction capability of our method, this paper analyzes the noise reduction performance of all the above-mentioned methods in the pipeline leakage state. The NPW signal at the time of the leak is collected, with the NPW signals from the upstream and downstream sensors denoted as A and B, respectively. The noise reduced signals are represented as A' and B', respectively. Taking the processing of A as an example, the optimal parameters obtained by our method are $[K, \alpha] = [13,2041]$, and the comparison with A' obtained by several other denoising methods is illustrated in Figure 14.

The Figure 14 (a) presents the original NPW signal. In Figure14 (b), the wavelet denoising suppresses most of the noise, and the pressure turning point of the denoised signal is clearer, but there is still overall sawtooth noise, especially at the 4s. From Figure 14(c), it can be observed that the signal after EMD processing is severely distorted. As shown in Figure 14 (d), the SNR-EMD noise reduction of the NPW signal has jagged noise near the turning point, which is unfavorable for leak location. The result of ADM-VMD processing is shown in Figure 14 (e), which works better than the other methods but still have some noise. From Figure 14 (f), it can be seen that signal after processing by the proposed method is overall smooth, the pressure fluctuation trend is preserved, and the pressure drop turning point at the time of leak is clear. The overall effect of our proposed method is better than ADM-VMD.

The quantitative assessment of the denoised signals A' obtained by various methods, utilizing the IE described in Chapter II, is presented in Table 3 (EMD is completely distorted and does not participate in the comparison). The IE values for A' obtained through wavelet, SNR-EMD and ADM-VMD are 2.303, 2.313 and 2.292, respectively. The IE value of A 'obtained by proposed method is 2.281, which is 0.131 lower than that of the original pressure signal compared with other methods. Therefore, the performance of the denoising method proposed in this paper is superior to the other four methods.

C. LEAK LOCATION

When a leak occurs in the pipeline, the NPW is generated inside the pipe, leading to a distinct sudden change in the NPW signal's time domain characteristics, referred to as the turning point. The turning points of pressure signals from sensors A and B are chosen as characteristic points for the time delay estimation method, and the time delay (Δt) between these turning points is calculated. Since leaks happen instantaneously, disrupting the steady state of the pipe, information about the leak is often reflected in the high-frequency components of the signal. After denoising the leakage induced negative pressure wave signal with the method proposed in this paper, VMD is used to decompose it, and the results are shown in Figure 15, where the blue arrows indicate the turning point and the three red arrows indicate the peak points of IMF3, IMF4 and IMF5.

The horizontal coordinates corresponding to the peak points of IMF3, IMF4 and IMF5 are consistent with the horizontal coordinates of the negative pressure wave turning points indicated by the blue arrows. This indicates that the leakage information is mainly contained in IMF3, IMF4 and IMF5. At the same time, in order to enhance the robustness of the method and avoid artificial selection of peak components, all high-frequency components except the lowest frequency components (IMF1-IMF5) were reconstructed. The reconstructed signal is the locating signal.

The denoised signal A' and its locating signal are shown in Figure 16(a), where the blue arrow points to its pressure

TABLE 4. Results of leak location.

L (m)	Leak point	L_1 (m)	Leak area (cm^2)	Wavelet			ADM-VMD			Proposed		
				L'_1 (m)	ε (m)	δ (%)	L'_1 (m)	ε (m)	δ (%)	L'_1 (m)	ε (m)	δ (%)
17.34	1	14.47	0.1	-	-	-	-	-	-	13.67	0.8	4.61
			0.2	-	-	-	17.67	3.2	18.45	14.67	0.2	1.15
	2	11.34	0.1	-	-	-	-	-	-	12.67	1.33	7.67
			0.2	4.67	6.67	38.47	6.67	4.67	26.93	11.67	0.33	1.9
	3	2.47	0.1	-	-	-	-	-	-	1.67	0.8	4.61
			0.2	-	-	-	-	-	-	2.67	0.2	1.15
23.34	4	17.34	0.1	-	-	-	14.67	2.67	11.44	16.67	0.67	2.87
			0.2	-	-	-	-	-	-	17.67	0.33	1.41
	5	14.47	0.1	-	-	-	-	-	-	13.67	0.8	3.43
			0.2	7.67	6.8	29.13	15.67	1.2	5.14	14.67	0.2	0.86
	6	2.47	0.1	-	-	-	-	-	-	3.67	1.2	5.14
			0.2	-	-	-	1.67	0.8	3.43	2.67	0.2	0.86
Average						53.04			13.08			2.97

turning point and the red arrow points to the peak point of the locating signal. The horizontal coordinate of the peak point, $t_1 = 10.356$s, means that a leak occurred in the pipe at the 10.356s when the upstream sensor A started collecting the pipe pressure signal.

The NPW signal collected by sensor B downstream of the pipe is processed in the same way. First, the denoised signal B' is obtained by using the proposed method ($[K, \alpha] = [11,2065]$ and the IE of B' is 2.3005). Then, the locating signal of B' is calculated using the method described in this subsection. Finally, the peak coordinates of the locating signal are obtained. Denoised signal B 'and its locating signal are shown in Figure 16(b). It can be seen from the figure that the horizontal coordinate of the peak point of the locating signal $t_2 = 10.350$s.

After t_1 and t_2 are obtained, it is possible to calculate $\Delta t = t_1 - t_2 = 10.356 - 10.350 = 0.006$s. Substituting $\Delta t = 0.006$s, $v = 1000$m/s and $L = 17.34$m into Eq (1) gives $L'_1 = 1/2(17.34 + 1000 \times 0.006) = 11.67$m, the distance between the leak point and the upstream sensor A is 11.67m. The actual distance between simulated leak point 2 and upstream sensor A is 11.34m. Therefore, the absolute error in leak location $\varepsilon = 0.33$m and the relative error $\delta = 1.9\%$.

In order to verify the effect of the method proposed in this paper, we will compare the locating results of the wavelet, EMD, SNR-EMD, ADM-VMD and the proposed method. Figure 17(a) shows the peak of the locating signal of the wavelet method. Δt obtained between two peak is -0.008s, and the absolute error of locating ε is 6.67m, and the relative errors δ is 38.47%, which is larger than the relative error of 1.9% of our method. In Figure 17(b), the EMD method could not find the signal crest and locating failed. As shown in Figure 17 (c), the locating signal of SNR-EMD have a peak, but due to the influence of noise, one of the peaks is the peak generated by noise, and the peak of the turning point is not correctly extracted, so the resulting Δt is large, resulting in location failure. The locating signal of ADM-VMD exhibits tangible peaks in Figure 17(d), with Δt is -0.004s and relative error δ is 26.93%, but still higher than the relative error of our method. As shown in Figure 17(e), the proposed method obtains the most prominent peak at the moment of leak.

In order to verify the performance of the algorithm, this paper conducted multiple leak location experiments. The NPW signals are processed using wavelet, EMD, SNR-EMD, ADM-VMD and our proposed method, as shown in Table 4. It is worth noting that the locating signals computed from the EMD denoised signals do not show peaks, and the location errors of SNR-EMD are large, and the results of both methods are not listed. The location absolute error is denoted by ε, and the relative error, defined as the ratio of ε to L, is represented by δ. If δ exceeds 50%, the location is considered unsuccessful, indicated by "-".

As can be seen from the locating results in Table 4, the signal denoised by wavelet successfully located the leak only 2 times, and ADM-VMD succeeded in 5 instances. The average relative error of wavelet is relatively high at 53.04%, while ADM-VMD denoising have an average relative error of 13.08%. The proposed exhibited a maximum relative error of 7.67%, a minimum relative error of 0.86%, and an average relative error of only 2.97% across the 12 experiments. Compared to other methods, the NPW signal location error after denoising with our method is smaller and more stable.

V. CONCLUSION

Noise in the pressure signal of a water pipeline can obscure signal characteristics, potentially impacting the accuracy of leak location. In this paper, we propose a novel method to address noise in pipeline pressure signals, aiming to enhance leak location accuracy. In summary, the key conclusions of this study are as follows:

1) The effectiveness of VMD in decomposing signals is significantly influenced by its parameter settings, particularly the number of IMFs, denoted as K, and the penalty factor, denoted as α. Appropriate parameter configurations result in improved VMD outcomes and enhanced decomposition efficiency. In contrast, the proposed method in the paper leverages Genetic Algorithm (GA), correlation coefficient (CC), and information entropy (IE) to globally optimize the

parameters K and α. This approach addresses the limitation of traditional methods, which often converge to local optimal parameters in VMD.

2) The proposed method demonstrates highly effective denoising capabilities. In the simulation study of signal denoising, wavelet, EMD, SNR-EMD, ADM-VMD and the proposed method are employed to denoise the signal individually. The proposed method achieves the highest SNR at 16.187dB, surpassing other methods and enhancing SNR by 9.387dB while preserving signal characteristics. In the study of NPW signal denoising in the pipeline, our method obtains a smooth noise-reduced signal with an IE of 2.281, reducing IE by 0.131 compared to the original pressure signal of 2.412 and outperforming other methods.

3) The NPW signal, after undergoing denoising with the proposed method, exhibits minimal error in leak location. Our proposed method demonstrates a maximum relative error of 7.67%, a minimum relative error of 0.86%, and an average relative error of only 2.97% across the 12 experiments. Compared with other signal processing methods, the NPW signal denoised by the proposed method has higher accuracy and stability in leak location.

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